InternNova Internship

Week 5 Assignment Report

Power BI & Business Intelligence

Name: Aryan Dubey

Task 1: Power BI Introduction & Data Import

Brief Explanation:

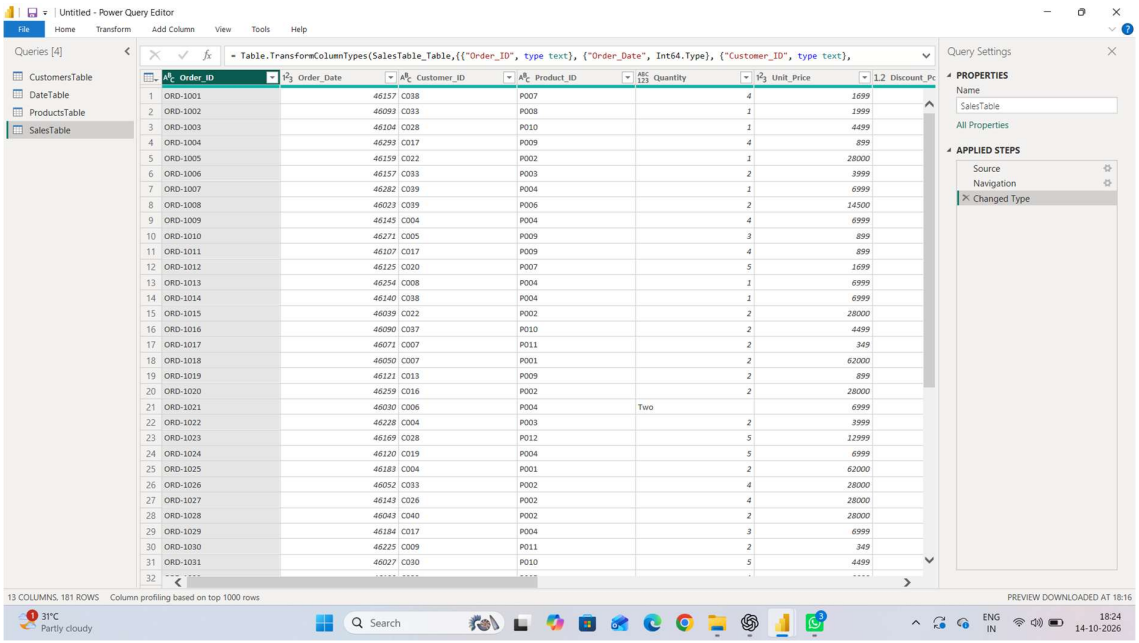
This task introduced the Power BI environment and the process of importing a multi-table Excel dataset. The dataset contains four related tables: SalesTable, ProductsTable, CustomersTable, and DateTable. The available columns and data types were reviewed in Power Query before analysis. Power BI is used in data analytics to import, transform, model, analyze, and visualize data so that business performance can be explored through interactive reports and dashboards.

Imported Tables:

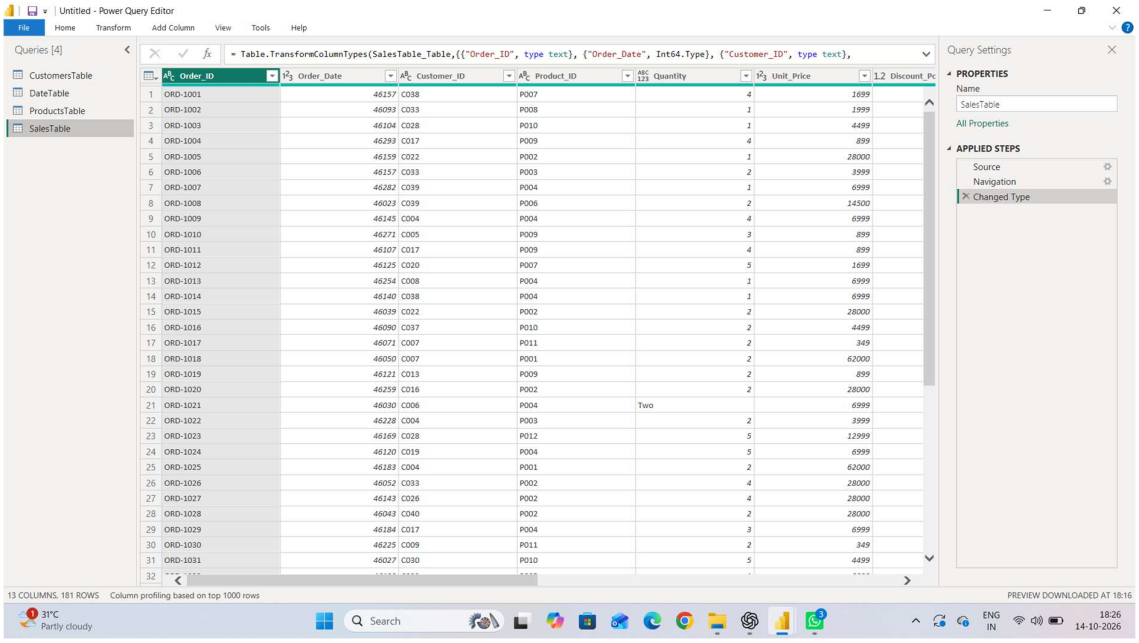
SalesTable stores transaction-level sales records. ProductsTable contains product details, CustomersTable contains customer attributes, and DateTable provides date fields for time-based analysis.

OUTPUT SCREENSHOT:

Imported Tables in Power Query



Columns and Data Types in SalesTable



	Order_ID	Order_Date	Customer_ID	Product_ID	Quantity	Unit_Price	Discount_Pc
1	ORD-1001	46157	C038	P007	4	1699	
2	ORD-1002	46093	C033	P008	1	1999	
3	ORD-1003	46104	C028	P010	1	4499	
4	ORD-1004	46293	C017	P009	4	899	
5	ORD-1005	46159	C022	P002	1	28000	
6	ORD-1006	46157	C033	P003	2	3999	
7	ORD-1007	46282	C039	P004	1	6999	
8	ORD-1008	46023	C039	P006	2	14500	
9	ORD-1009	46145	C004	P004	4	6999	
10	ORD-1010	46271	C005	P009	3	899	
11	ORD-1011	46107	C017	P009	4	899	
12	ORD-1012	46125	C020	P007	5	1699	
13	ORD-1013	46254	C008	P004	1	6999	
14	ORD-1014	46140	C038	P004	1	6999	
15	ORD-1015	46039	C022	P002	2	28000	
16	ORD-1016	46090	C037	P010	2	4499	
17	ORD-1017	46071	C007	P011	2	349	
18	ORD-1018	46050	C007	P001	2	62000	
19	ORD-1019	46121	C013	P009	2	899	
20	ORD-1020	46259	C016	P002	2	28000	
21	ORD-1021	46030	C006	P004	Two	6999	
22	ORD-1022	46228	C004	P003	2	3999	
23	ORD-1023	46169	C028	P012	5	12999	
24	ORD-1024	46120	C019	P004	5	6999	
25	ORD-1025	46183	C004	P001	2	62000	
26	ORD-1026	46052	C033	P002	4	28000	
27	ORD-1027	46143	C026	P002	4	28000	
28	ORD-1028	46043	C040	P002	2	28000	
29	ORD-1029	46184	C017	P004	3	6999	
30	ORD-1030	46225	C009	P011	2	349	
31	ORD-1031	46027	C030	P010	5	4499	
32							

Task 2: Data Cleaning & Preparation

Brief Explanation:

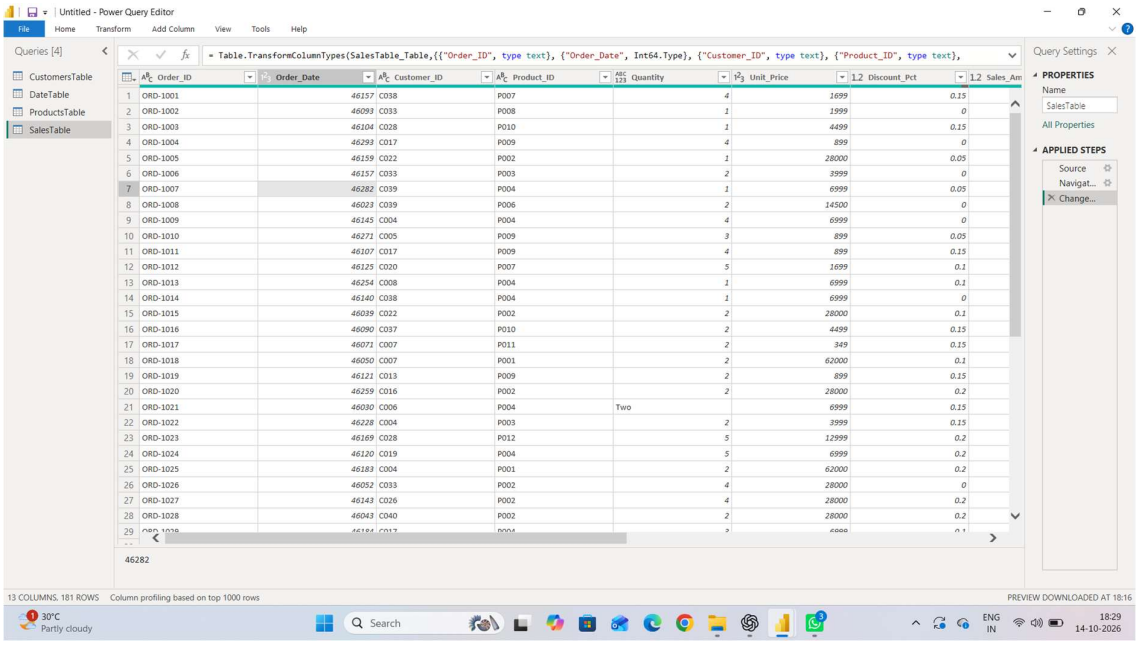
Power Query was used to inspect and prepare the dataset before modeling. The cleaning process included correcting data types, handling an invalid Quantity value, removing an unnecessary Notes column, removing duplicate records, standardizing inconsistent text values, reviewing customer-age issues, and verifying the Products and Date tables. These transformations improved consistency and prepared the tables for reliable analysis.

Main Transformations:

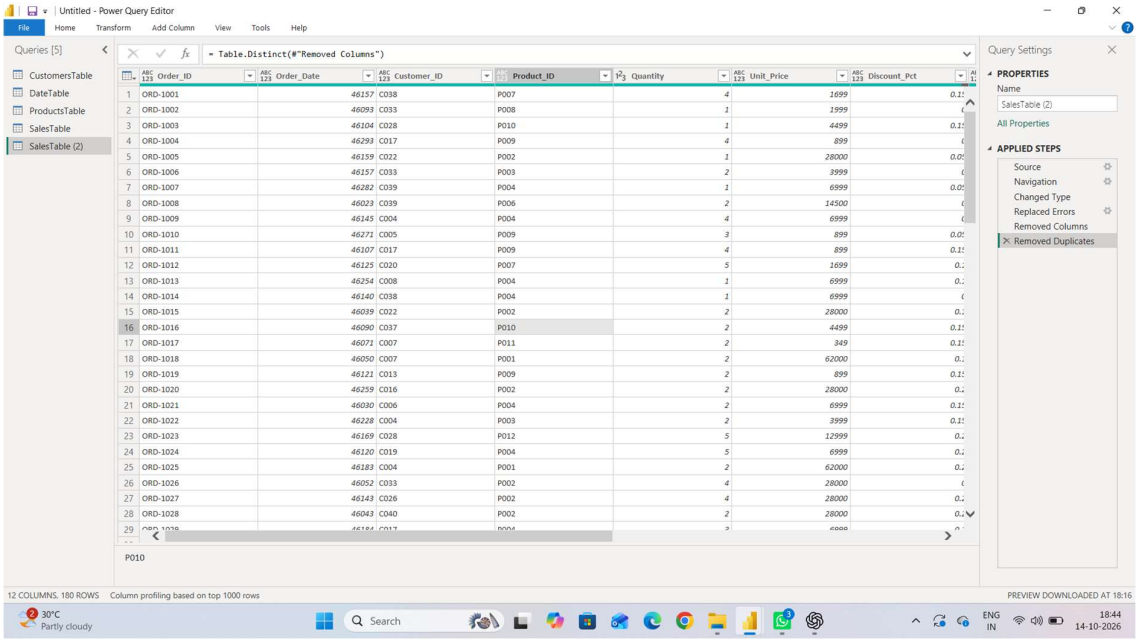
• SalesTable: Quantity was converted to Whole Number, the resulting error from the text value 'Two' was replaced with 2, the unnecessary Notes column was removed, and duplicate rows were removed.
• CustomersTable: customer fields were inspected, invalid/missing values were handled during cleaning, region text was standardized, and duplicate customer records were removed.
• ProductsTable: category text was standardized and numeric columns were verified.
• DateTable: Date, Year, Month_No, Month_Name, and Quarter data types were checked and corrected where required.

OUTPUT SCREENSHOT:

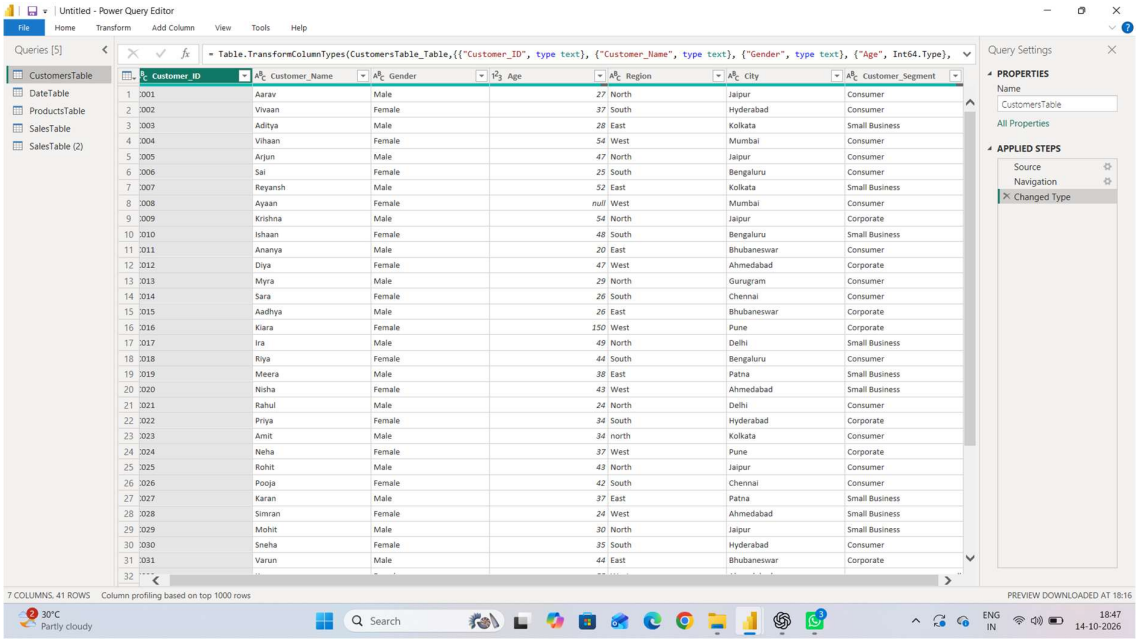
SalesTable - Before Cleaning



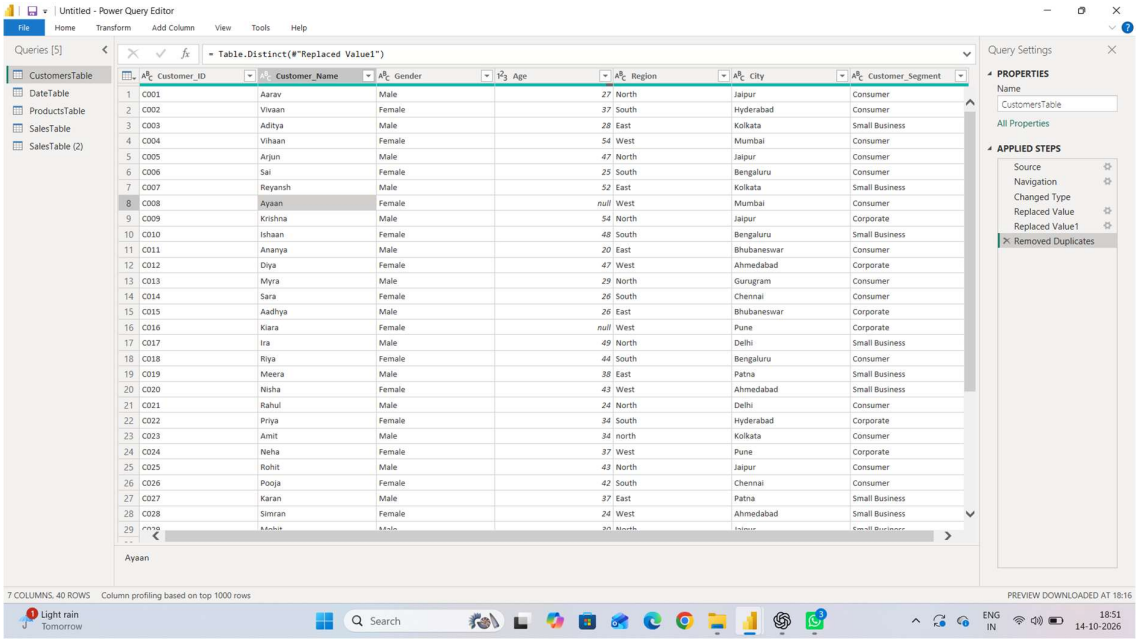
SalesTable - After Cleaning



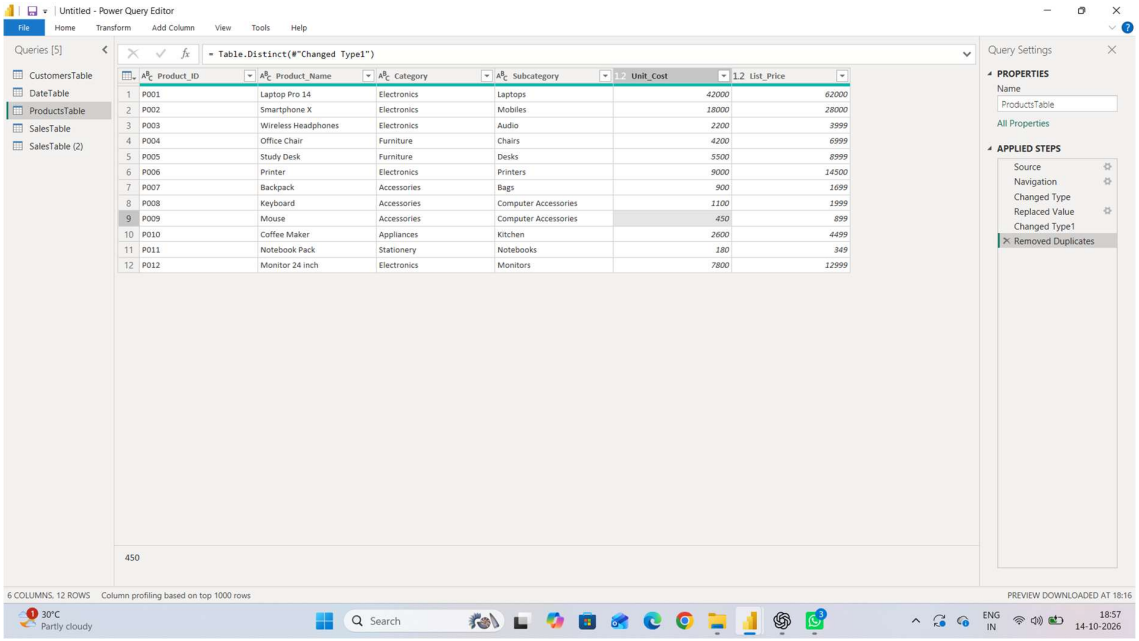
CustomersTable - Before Cleaning



CustomersTable - Cleaning Steps and Duplicate Removal



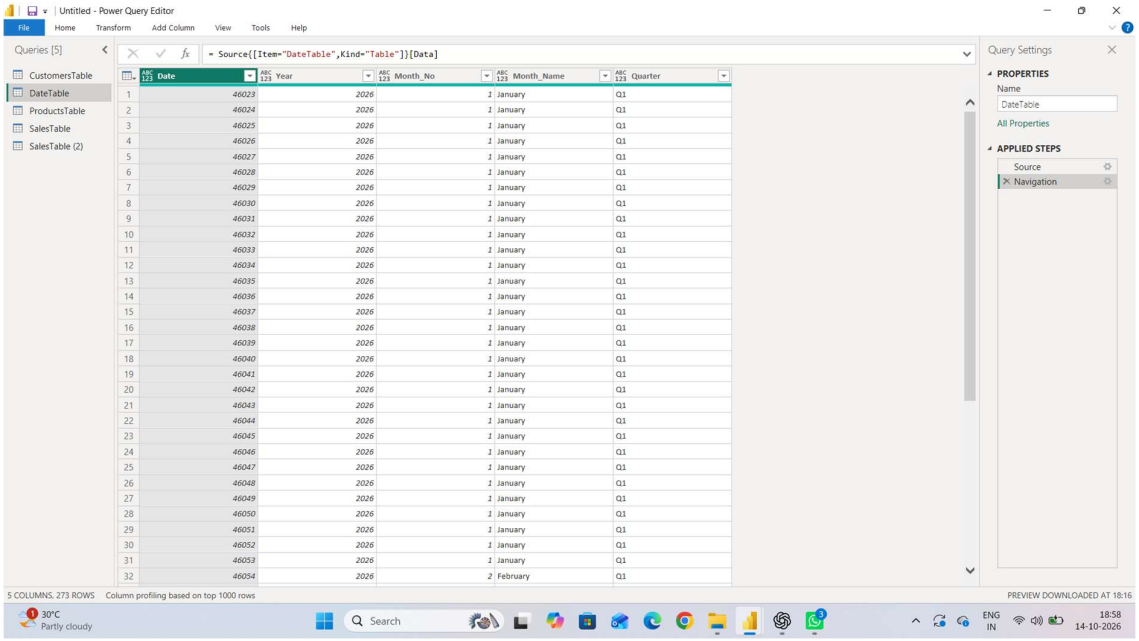
ProductsTable - After Cleaning



	Product_ID	Product_Name	Category	Subcategory	Unit_Cost	List_Price
1	P001	Laptop Pro 14	Electronics	Laptops	42000	62000
2	P002	Smartphone X	Electronics	Mobiles	18000	28000
3	P003	Wireless Headphones	Electronics	Audio	2200	3999
4	P004	Office Chair	Furniture	Chairs	4200	6999
5	P005	Study Desk	Furniture	Desks	5500	8999
6	P006	Printer	Electronics	Printers	9000	14500
7	P007	Backpack	Accessories	Bags	900	1699
8	P008	Keyboard	Accessories	Computer Accessories	1100	1999
9	P009	Mouse	Accessories	Computer Accessories	450	899
10	P010	Coffee Maker	Appliances	Kitchen	2600	4499
11	P011	Notebook Pack	Stationery	Notebooks	180	349
12	P012	Monitor 24 inch	Electronics	Monitors	7800	12999

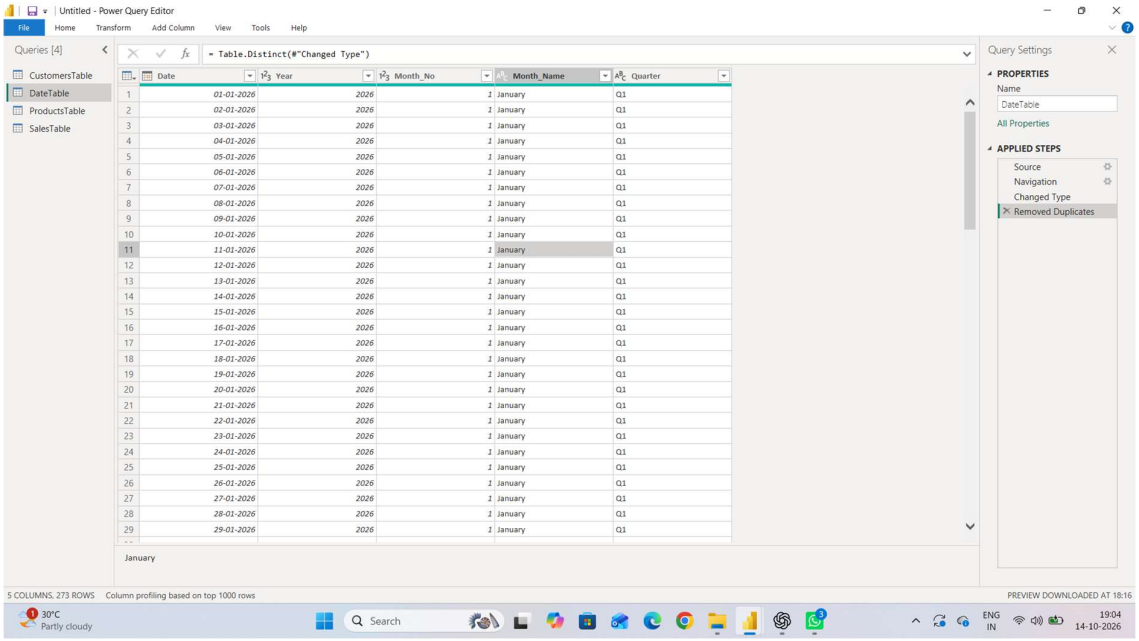
450

DateTable - Before Data Type Correction



	Date	Year	Month_No	Month_Name	Quarter
1	46023	2026	1	January	Q1
2	46024	2026	1	January	Q1
3	46025	2026	1	January	Q1
4	46026	2026	1	January	Q1
5	46027	2026	1	January	Q1
6	46028	2026	1	January	Q1
7	46029	2026	1	January	Q1
8	46030	2026	1	January	Q1
9	46031	2026	1	January	Q1
10	46032	2026	1	January	Q1
11	46033	2026	1	January	Q1
12	46034	2026	1	January	Q1
13	46035	2026	1	January	Q1
14	46036	2026	1	January	Q1
15	46037	2026	1	January	Q1
16	46038	2026	1	January	Q1
17	46039	2026	1	January	Q1
18	46040	2026	1	January	Q1
19	46041	2026	1	January	Q1
20	46042	2026	1	January	Q1
21	46043	2026	1	January	Q1
22	46044	2026	1	January	Q1
23	46045	2026	1	January	Q1
24	46046	2026	1	January	Q1
25	46047	2026	1	January	Q1
26	46048	2026	1	January	Q1
27	46049	2026	1	January	Q1
28	46050	2026	1	January	Q1
29	46051	2026	1	January	Q1
30	46052	2026	1	January	Q1
31	46053	2026	1	January	Q1
32	46054	2026	2	February	Q1

DateTable - After Data Type Verification



Task 3: Data Modeling & Relationships

Brief Explanation:

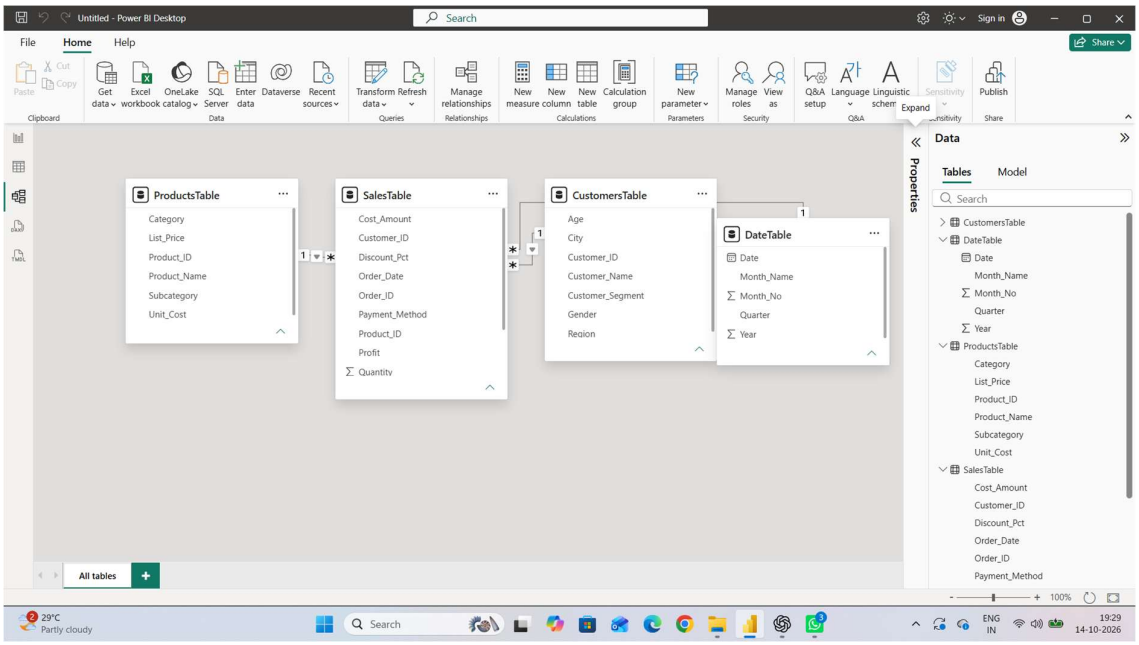
A basic star-schema style data model was created in Power BI. SalesTable acts as the fact table because it stores transaction-level sales records. ProductsTable, CustomersTable, and DateTable act as dimension tables because they provide descriptive attributes used to filter and analyze sales. The relationships use common key columns and follow one-to-many cardinality from each dimension table to SalesTable.

Relationships Created:

ProductsTable[Product_ID] (1) → SalesTable[Product_ID] (*)
CustomersTable[Customer_ID] (1) → SalesTable[Customer_ID] (*)
DateTable[Date] (1) → SalesTable[Order_Date] (*)

OUTPUT SCREENSHOT:

Power BI Data Model and Relationships



Task 4: DAX Calculated Measures

Brief Explanation:

DAX measures were created to summarize business performance dynamically. Measures calculate results based on the current filter context, which means the values can respond to slicers and report interactions. Seven measures were created, exceeding the minimum requirement of three.

DAX Measures:

```
Total Sales = SUM(SalesTable[Sales_Amount])

Average Sales = AVERAGE(SalesTable[Sales_Amount])

Total Orders = DISTINCTCOUNT(SalesTable[Order_ID])

Total Profit = SUM(SalesTable[Profit])

Maximum Sales = MAX(SalesTable[Sales_Amount])

Minimum Sales = MIN(SalesTable[Sales_Amount])

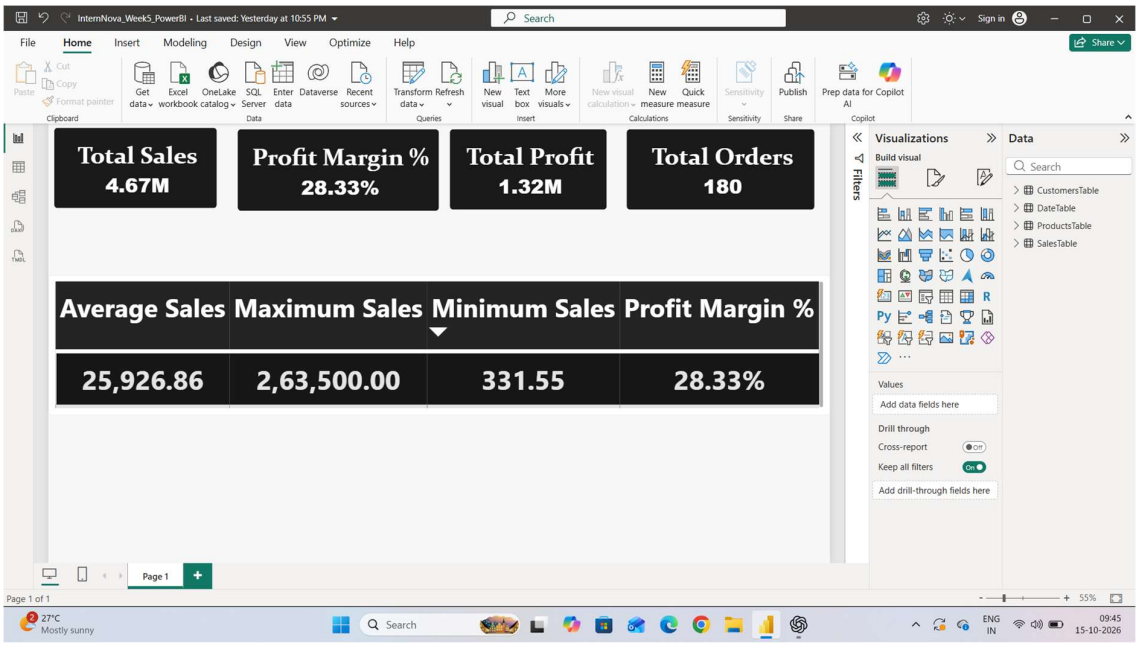
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Measure Results:

Total Sales ≈ 4.67M, Total Profit ≈ 1.32M, Total Orders = 180, Profit Margin ≈ 28.33%, Average Sales ≈ 25,926.86, Maximum Sales = 263,500.00, and Minimum Sales = 331.55.

OUTPUT SCREENSHOT:

DAX Measure Results



Task 5: Power BI Visualizations

Brief Explanation:

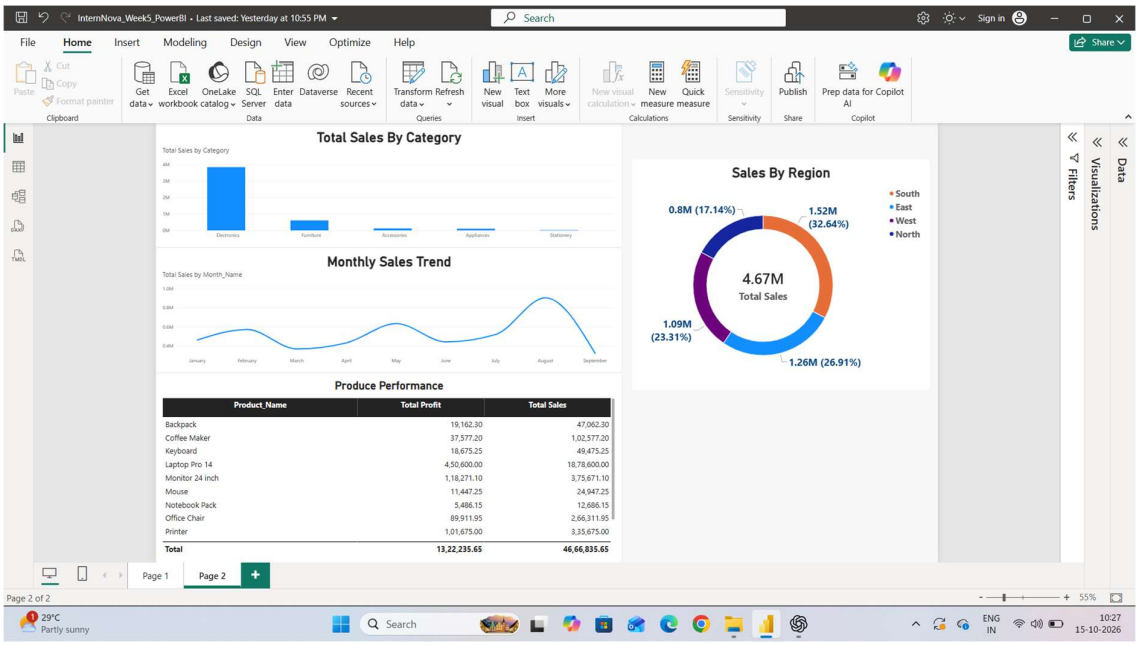
Meaningful Power BI visuals were created to present business performance from different perspectives. The report page includes a clustered column chart for category performance, a line chart for monthly trends, a donut chart for regional contribution, and a product-performance table. KPI/Card visuals were also created in Task 4.

Visualizations Created:

- Clustered Column Chart – Total Sales by Category
- Line Chart – Monthly Sales Trend
- Donut Chart – Sales by Region
- Table – Product Performance using Product Name, Total Sales, and Total Profit
- KPI/Card – Total Sales, Total Profit, Total Orders, and Profit Margin

OUTPUT SCREENSHOT:

Power BI Visualization Page



Task 6: Interactive Dashboard & Filters

Brief Explanation:

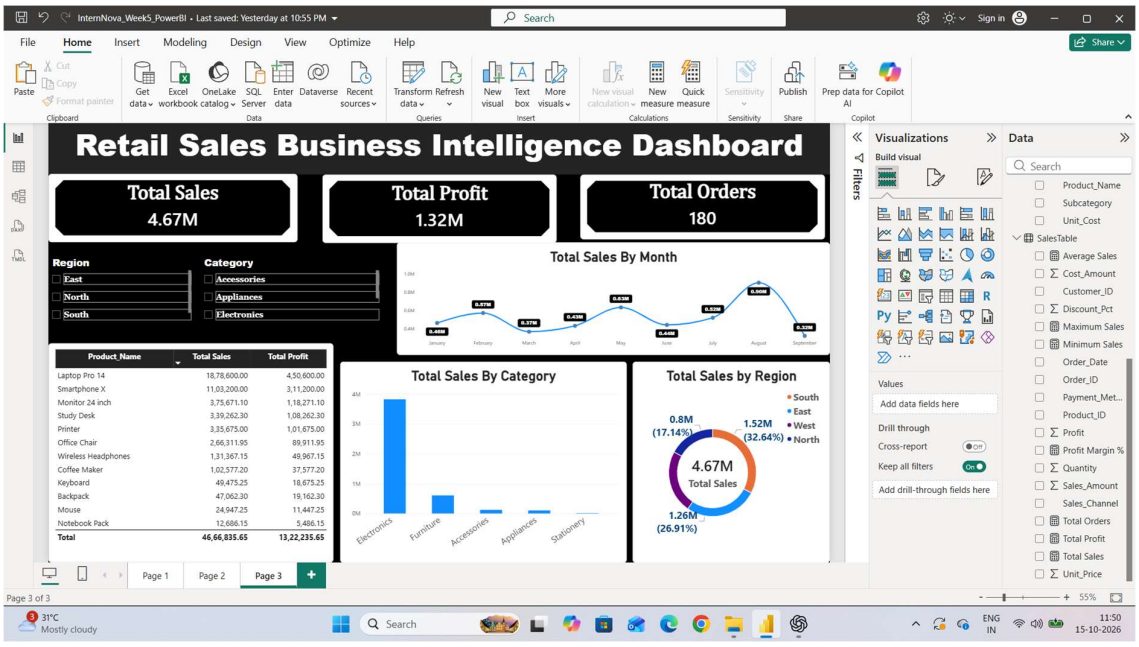
An interactive Retail Sales Business Intelligence Dashboard was developed. The dashboard contains three KPI cards, multiple business visuals, and two slicers. Region and Category slicers allow users to filter the dashboard and explore sales performance interactively. The layout was organized so that key metrics are visible first, followed by trends, category performance, regional performance, and product-level results.

Dashboard Components:

- KPI Cards: Total Sales, Total Profit, Total Orders
- Slicers: Region and Category
- Line Chart: Total Sales by Month
- Column Chart: Total Sales by Category
- Donut Chart: Total Sales by Region
- Product Performance Table

OUTPUT SCREENSHOT:

Interactive Retail Sales Business Intelligence Dashboard



Task 7: Business Insights & Recommendations

Brief Explanation:

The dashboard was analyzed to identify important trends, top and low performers, and opportunities for business action. The insights below are based on the final Power BI results.

Business Insights:

1. Electronics is the highest-performing category, generating the largest share of total sales and significantly outperforming the other categories.

2. August recorded the highest monthly sales at approximately 0.90M, while September recorded the lowest at approximately 0.32M.

3. South is the highest-performing region, contributing approximately 1.52M (32.64%) of total sales. East follows with approximately 1.26M (26.91%).

4. Laptop Pro 14 is the highest-performing product with approximately 1.88M in sales, followed by Smartphone X at approximately 1.10M.

5. The business generated approximately 4.67M in total sales and 1.32M in total profit, producing a profit margin of about 28.33% across 180 orders.

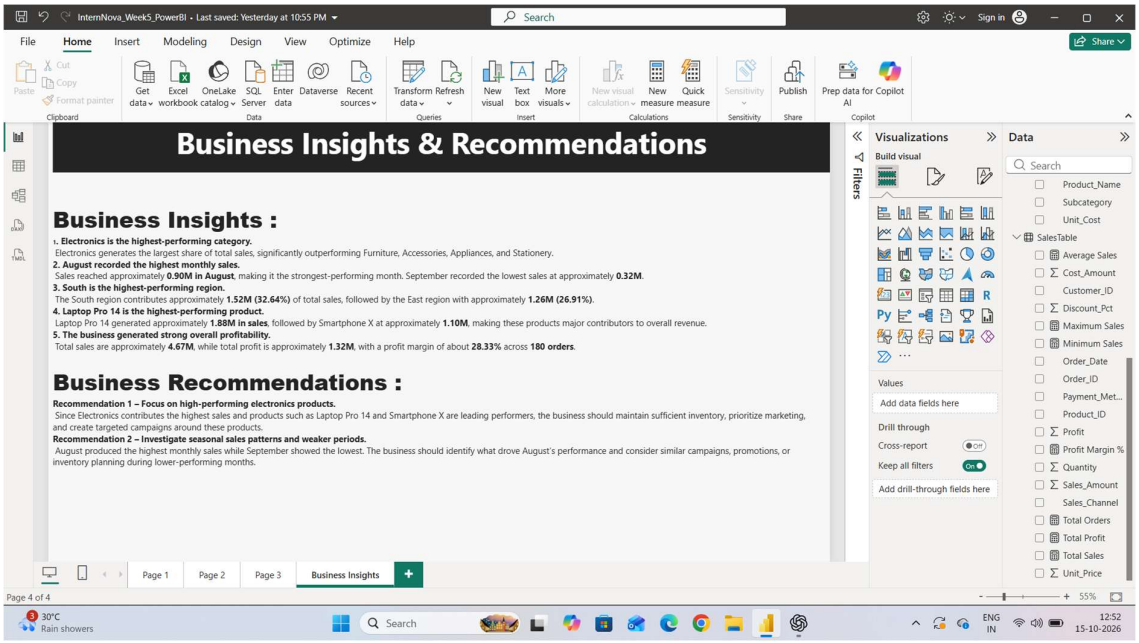
Business Recommendations:

1. Focus on high-performing electronics products. Maintain sufficient inventory, prioritize marketing, and create targeted campaigns around leading products such as Laptop Pro 14 and Smartphone X.

2. Investigate seasonal sales patterns and weaker periods. August was the strongest month while September was the weakest, so the business should identify the factors behind August's performance and apply suitable campaigns, promotions, or inventory planning during lower-performing months.

OUTPUT SCREENSHOT:

Business Insights and Recommendations



GOOGLE DRIVE :

https://drive.google.com/drive/folders/1hi3NZRiaWAd8Qol0pSfw8Fat1NxiP1n9?usp=sharing